



Required Capstone Project: Module 6

Due Jul 8 by 11:59pm**Points** None**Available** after Jul 1 at 12am

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**Learning Outcomes Addressed**

- Understand how AI, GenAI and Agentic AI work and where they apply in business.
- Use prompts effectively for analysis, decision-making and content generation.
- Map business workflows and identify where AI can drive automation and efficiency.
- Evaluate AI opportunities using feasibility, ROI and impact criteria.
- Build AI adoption roadmaps aligned to business goals and KPIs.
- Apply data governance, privacy and ethical AI principles.

Phase 3 – Agentic AI Design & Control

Milestone 6: Agent Architecture Blueprint

Estimated Effort: 1.5 Hours

Objective

In the previous modules of the capstone, you explored how Generative AI can improve and transform business workflows.

Specifically:

- In **Module 1**, you selected a business workflow and defined an **AI opportunity**.
- In **Module 2**, you analysed the **current workflow and operational risks**.
- In **Module 3**, you designed **prompts that enable Generative AI to perform workflow tasks**.
- In **Module 4**, you estimated **efficiency and financial impact**.
- In **Module 5**, you redesigned the workflow to reflect an **AI-enabled future state**.

In this module, you will move one step further by exploring **Agentic AI systems**.

Unlike simple AI prompts, **AI agents can plan actions, use tools, retain context, and execute multi-step workflows autonomously**.

You will design a **conceptual Agent Architecture Blueprint** showing how an AI agent could support the workflow you redesigned in the previous module.

This blueprint will outline how the agent would:

- Understand its objective
- Plan actions
- Interact with external tools

- Maintain memory or context
- Operate with an appropriate level of autonomy

The goal is to demonstrate how **agent-based systems can coordinate complex business workflows**.

Note: This is a required assignment and counts towards programme completion.

Task Overview

Design a **conceptual architecture for an AI agent** that supports the business workflow you analysed and redesigned earlier in the capstone.

Your design should define the following components:

- The **agent's objective**
- The **planning workflow** the agent follows
- The **tools or systems the agent interacts with**
- The **memory structure used by the agent**
- The **level of autonomy assigned to the agent**
- The **estimated cost of running the agent per task or transaction**

Your blueprint should explain **how the agent operates inside the workflow** rather than focusing on technical implementation details.

Step 1 — Define the Agent Objective

Start by defining the **primary goal of the AI agent**.

The objective should clearly explain:

- What task or workflow the agent supports
- What outcome it is responsible for producing
- What business value it delivers

The objective should be **specific and measurable**.

Example (Illustrative)

Workflow: Customer support request handling

Agent Objective:

Automatically analyse incoming customer support requests, classify the issue, generate a draft response using the knowledge base, and route the case to a support agent for final approval.

Business Value:

Reduce response time and improve support team productivity.

Step 2 — Define the Agent Planning Workflow

AI agents typically perform tasks using **multi-step reasoning and planning**.

Define the sequence of actions the agent would follow to complete its objective.

Your planning workflow should show:

- How the agent receives input
 - How it analyses the information
 - How it decides what action to take
 - How it produces outputs
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Example (Illustrative)

Customer Support Agent Planning Workflow

1. Receive incoming customer support request
2. Analyse the request using a language model
3. Classify the issue category
4. Retrieve relevant knowledge base articles
5. Generate a draft response
6. Send the response to a human agent for approval
7. Deliver the approved response to the customer

This step-by-step plan describes **how the agent completes the workflow**.

Step 3 — Define Tool Integrations

AI agents often interact with **external tools and enterprise systems**.

These tools allow the agent to retrieve information, update records, or trigger actions.

Identify the systems your agent would need to interact with.

Examples include:

- CRM systems
- Internal databases
- Document repositories
- Analytics platforms
- Email or messaging systems

Example (Illustrative)

Tools Used by the Agent

Tool	Purpose
CRM system	Retrieve customer account information
Knowledge base	Search relevant help articles
Email platform	Send responses to customers
Ticketing system	Update support ticket status

These integrations enable the agent to **operate inside real business workflows**.

Step 4 — Design the Agent Memory Structure

AI agents often require memory to retain context across interactions.

Define whether your agent requires:

- **Short-term memory** (context within a single task)
- **Long-term memory** (knowledge retained across tasks)

Memory design helps determine **how much context the agent can maintain**.

Example (Illustrative)

Short-Term Memory

Stores the current customer query and conversation history.

Long-Term Memory

Stores past support interactions to help identify recurring issues.

Not all agents require long-term memory, but it can enable **more intelligent decision-making**.

Step 5 — Define the Autonomy Level

AI agents can operate with different levels of autonomy.

Define how much decision-making authority the agent should have.

Common Autonomy Levels

Assistive Agent

AI generates suggestions or drafts.

Humans make final decisions.

Example:

AI drafts customer responses, but support agents approve them.

Semi-Autonomous Agent

AI performs actions but requires occasional human validation.

Example:

AI automatically categorises tickets but escalates complex cases to humans.

Fully Autonomous Agent

AI executes tasks without human approval.

Example:

AI automatically processes simple refund requests.

For most enterprise use cases, **assistive or semi-autonomous models are more common**.

Step 6 — Estimate Execution Cost

Finally, estimate the **cost of running the agent per task or transaction**.

This may include:

- AI model inference cost
- API usage costs
- Cloud computing resources
- Data retrieval costs

You may use a **simplified estimate**.

Example (Illustrative)

Estimated Cost per Execution

Component	Cost Estimate
AI model inference	\$0.03
Knowledge base search	\$0.01
API calls	\$0.02

Estimated total cost per support request:

\$0.06 per execution

These estimates help organisations evaluate **the economic viability of agent-based systems**.

Required Output Structure

Your **Agent Architecture Blueprint** should include the following sections:

Agent Objective and Business Context

Describe the purpose of the agent and the workflow it supports.

Agent Planning Workflow

Explain the step-by-step actions the agent performs.

Tool Ecosystem

List the tools, systems, or APIs the agent interacts with.

Memory Design

Describe the short-term and long-term memory used by the agent (if applicable).

Autonomy Level and Human Oversight

Explain how much control the agent has and where humans remain involved.

Estimated Execution Cost

Provide a simplified estimate of cost per task or transaction.

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Practical Tips

Keep the Design Conceptual

This exercise focuses on **agent architecture thinking**, not detailed technical implementation.

Focus on how the agent **operates within the workflow**.

Connect to Earlier Modules

Your agent should ideally support the workflow redesigned in **Module 5**.

The tasks performed by the agent may relate to:

- Prompts designed in **Module 3**
 - Efficiency improvements identified in **Module 4**
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Focus on Human–AI Collaboration

Most enterprise agents operate alongside humans.

Consider where **human oversight or approval** should remain in the workflow.

Avoid These Common Mistakes

- ✗ Designing agents that attempt to perform every workflow step
- ✗ Ignoring the need for tool integrations

✗ Assigning full autonomy without governance or oversight

✗ Designing agents that operate without clear objectives

Effective agents have **clear goals, structured planning, and defined boundaries.**

Submission

Before submitting your **Agent Architecture Blueprint**, confirm that you have included:

- ✓ Agent objective and business context
- ✓ Step-by-step planning workflow
- ✓ Tool integrations used by the agent
- ✓ Memory design (short-term / long-term if applicable)
- ✓ Autonomy level and human involvement points
- ✓ Estimated cost per execution or transaction

Instructions:

- Use the same document that you downloaded in Module 1 and include your responses.
- We recommend you add additional text boxes or slides (if needed) for each milestone task to ensure that your responses are documented in a proper format.
- The document will gradually grow as you complete each capstone milestone. The final submission occurs at the end of Module 14.

Grading Rubric

Maximum Points: 15

Criteria	Full Points	0 Points
Agent Objective & Workflow Design (4 points)	The submission defines the agent objective, business context, and workflow supported by the agent. The workflow steps are clearly presented. (4 points)	Agent objective, business context, or workflow design is missing. (0 points)
Planning Workflow, Tools & Memory Design (4 points)	The submission includes a step-by-step planning workflow, relevant tool integrations, and a memory design. The role of each component is explained. (4 points)	Planning workflow, tool integrations, or memory design are missing. (0 points)
Autonomy Level & Human Oversight (4 points)	The submission defines the agent's autonomy level, identifies human involvement points, and explains governance or oversight mechanisms (4 points)	Autonomy level, human oversight, or governance considerations are missing. (0 points)
Execution Cost Estimation (3 points)	The submission includes a estimate of execution cost per task or transaction. Cost assumptions and calculations are presented. (3 points)	Execution cost estimates or calculations are missing. (0 points)

Why this Step Matters

Agentic AI represents the next stage of Generative AI adoption.

Instead of responding to individual prompts, AI systems can:

- Plan tasks
- Coordinate tools

- Execute workflows
- Collaborate with humans

Understanding how to design **agent architectures** helps organisations move from simple AI experiments to **scalable intelligent automation systems**.

In the next module, you will explore how to **govern, monitor, and control agentic AI systems to ensure safe and reliable operation**.